



Datta Meghe College of Engineering

Airoli, Navi Mumbai

Department of Information Technology

University of Mumbai
Academic Session : 2023 - 24

A Presentation
On
“QuickPrint: A Web-Based Printing Platform for Academic Communities”

Presented By

Vaishnavi Jamkhandi	B.E., Semester 7, Div A/B, Roll No.- 50
Isha Jogmarge	B.E., Semester 7, Div A/B, Roll No.- 53
Dhruv Kadam	B.E., Semester 7, Div A/B, Roll No.- 55
Prachi Kadam	B.E., Semester 7, Div A/B, Roll No.- 56

Under the Guidance of
Prof. Jayashree Phonde

Content

- **Introduction**
- **Literature Review**
- **Problem Definition**
- **Aims and Objectives**
- **Proposed Methodology**
- **Time Frame**
- **References**

Introduction

- QuickPrint is a smart, full-stack printing platform designed to simplify the process of document uploading, printing, payment, and order tracking.
- It connects customers, print-shop vendors, and administrators through a single digital platform.
- Customers can upload documents, configure printing requirements, place orders, make online payments, and track their orders.
- Vendors can manage print queues, orders, printers, and shop operations, while administrators manage the overall system.
- The existing QuickPrint project serves as the base system for this BE Open Source Project.
- The project aims to enhance the existing system into a secure, modular, scalable, and community-contributable open-source platform.

Literature Review

Research Gap:

Existing studies mainly address individual aspects such as printing services, queue management, booking, billing, tracking, scalability and responsive design.

QuickPrint integrates these aspects into a single platform connecting customers, vendors, and administrators, with payment processing and intelligent queue/wait-time management.

Problem Definition

Problem Definition: The existing QuickPrint platform provides a digital workflow for document upload, printing, payment and order management, but several limitations restrict its security, scalability, reliability, and extensibility.

Key Problems:

- Security limitations — inadequate password protection, exposed/debug APIs, and absence of proper rate limiting.
- Monolithic architecture — a large backend makes maintenance and contribution difficult.
- Limited order management — no proper order cancellation and refund workflow.
- Payment reliability — limited handling and testing of payment failure and transaction scenarios.
- Lack of notifications — no complete customer notification workflow for order status updates.
- Limited ML integration — some wait-time prediction inputs are hardcoded instead of using actual system data.
- Usability limitations — improvements are required for responsive and multi-file workflows.
- Limited testing & documentation — greater testing and open-source documentation are required for community contribution.

Aims And Objectives

Aim: To transform the existing QuickPrint project into a secure, modular, scalable, and community-contributable open-source smart printing platform.

Objectives:

1. Enhance Security

Implement secure authentication, password protection, RBAC, API protection, and rate limiting.

2. Improve User Experience

Provide responsive UI, multiple file uploads, improved document preview, and better order management.

3. Improve Payment & Order Handling

Introduce cancellation/refund workflows and systematic Razorpay testing.

4. Introduce Notifications

Provide email-based updates for important order events.

5. Improve ML Integration

Replace hardcoded inputs with actual system data for better wait-time prediction.

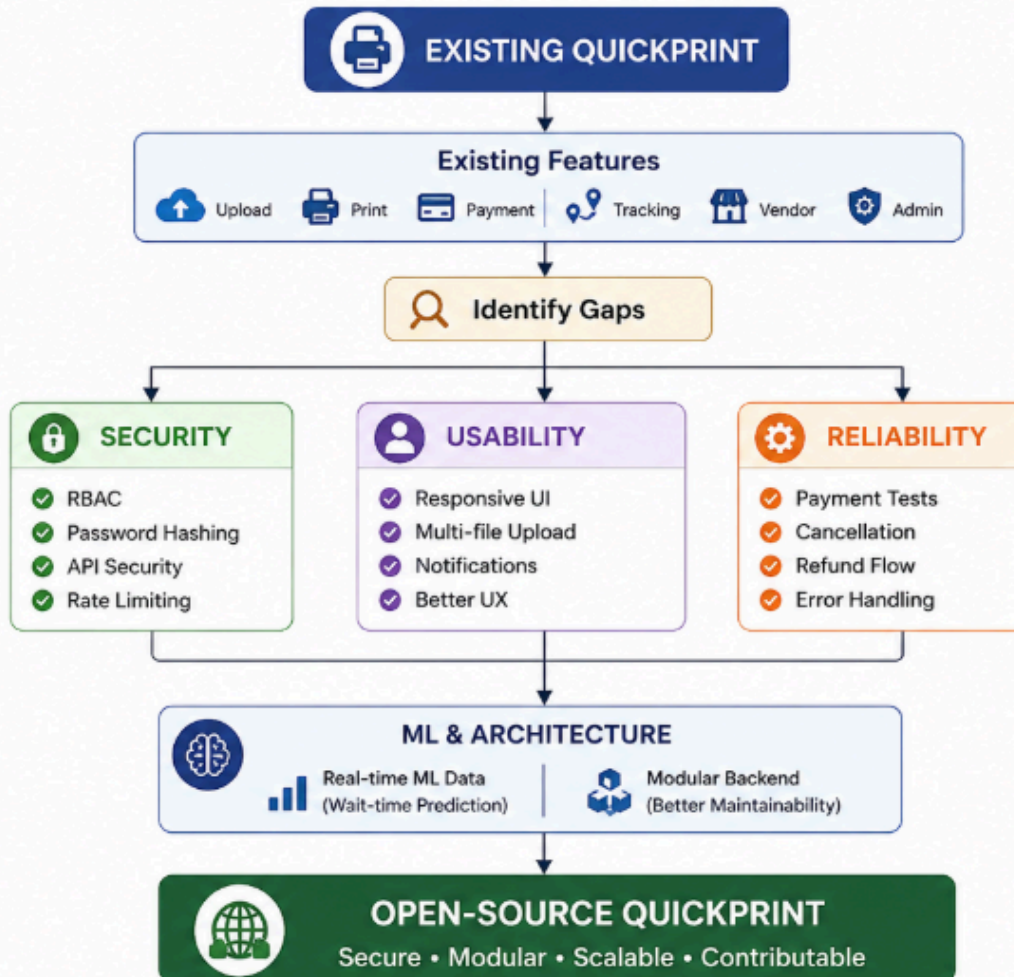
6. Improve Maintainability

Modularize the existing backend to make the code easier to understand, maintain, and extend.

7. Enable Open-Source Collaboration

Provide GitHub documentation, contribution guidelines, issue tracking and a suitable open-source license.

Proposed Methodology



MAJOR PROPOSED ENHANCEMENTS	
 Security	Password hashing, RBAC, API protection, rate limiting
 Customer	Cancellation, multiple file upload, responsive UI
 Payment	Failure testing, cancellation/refund handling
 Notifications	Email-based order updates
 Vendor	Improved queue and printing workflow
 ML	Replace hardcoded inputs with actual system data
 Architecture	Modularize backend for easier maintenance
 Open Source	GitHub, LICENSE, README, CONTRIBUTING & issue/PR workflow

Proposed Changes: What, Where & How

Change	Where?	How?
 Security & RBAC	Authentication, Backend APIs	Hash passwords, implement role-based middleware, protect APIs, add rate limiting
 Order Cancellation	Customer + Order Module	Add cancellation option, validate order status, store reason/time, update status
 Refund Handling	Payment + Order Module	Connect cancellation with payment status and implement refund workflow
 Multiple File Upload	Customer / Upload Module	Allow multiple documents, process each file, calculate combined page count
 Responsive UI	Customer, Vendor & Admin UI	Improve CSS layouts, tables, dashboards and navigation for mobile/tablet
 Email Notifications	Backend / Order Events	Trigger emails for order confirmation, printing and ready-for-pickup events
 Razorpay Testing	Payment Module	Test success, failure, cancellation, invalid signature and duplicate callbacks
 Admin Analytics	Admin Dashboard	Add order, revenue, vendor and cancellation statistics with charts
 ML Improvement	Wait-Time Prediction	Replace hardcoded inputs with actual printer, queue and order data
 Backend Modularization	Existing Backend	Separate large backend logic into routes, controllers and services
 Open-Source Setup	GitHub / Documentation	Add README, LICENSE, CONTRIBUTING, issue templates and PR workflow

Time Frame (Planning) of completion of project

Activity	W1	W2	W3	W4	W5	W6	W7	W8
 Study & Analysis								
 Open-Source Setup								
 Security & Core Enhancements								
 Payment & Notifications								
 ML Improvements								
 Architecture & UI Enhancement								
 Testing & Validation								
 Documentation & Release								

References

- [1] A. Sharma, “Online printing shop: A digital approach to printing services,” *Int. J. Comput. Appl.*, vol. 42, no. 3, pp. 112–118, Mar. 2020.
- [2] R. Patel and S. Mehta, “PrintFlow: Queue management system for cloud-based printing services,” in *Proc. Int. Conf. Inf. Commun. Technol.*, 2019, pp. 234–239.
- [3] K. Rao and M. Deshpande, “Web-based slot booking system for academic printing services,” in *Proc. IEEE Int. Conf. Adv. Comput., Commun. Control Syst.*, 2021, pp. 1–6.
- [4] M. Buyya, C. Vecchiola, and S. T. Selvi, *Mastering Cloud Computing*. New Delhi: McGraw Hill, 2019.
- [5] S. Basu, *Platform Scalability in Digital Services*. New York: Springer, 2020.
- [6] J. Lee and H. Kim, “Automated billing systems for web platforms,” *IEEE Access*, vol. 6, pp. 45231–45240, 2018.
- [7] P. Kumar and R. Sharma, “E-invoicing systems for small and medium enterprises,” *Int. J. Econ. Bus. Manag.*, vol. 9, no. 2, pp. 45–52, Jun. 2021.

References

- [8] C. Liu, “Queue management systems in public services,” *ACM Comput. Surv.*, vol. 50, no. 4, pp. 1–35, Nov. 2017.
- [9] D. Banerjee and S. Roy, “Smart queuing frameworks for service efficiency,” *Procedia Comput. Sci.*, vol. 171, pp. 1234–1241, 2020.
- [10] H. Singh, “Mobile-first design principles for responsive web applications,” in *Mobile Web Design: Principles and Practice*. Berlin: Springer, 2021, pp. 89–115.
- [11] T. Nguyen, “Cloud queue monitoring and analytics,” in *Proc. IEEE Int. Conf. Cloud Comput.*, 2020, pp. 567–572.
- [12] L. Chen and F. Wang, “Improving user satisfaction through live order tracking systems,” *IEEE Trans. Serv. Comput.*, vol. 15, no. 2, pp. 890–901, Mar. 2022.
- [13] F. Ahmed, “Digital platforms for academic services: A case study,” *Int. J. Comput. Sci. Inf. Technol.*, vol. 11, no. 4, pp. 23–30, Aug. 2019.
- [14] P. Mehta, “File handling and compression techniques in web-based systems,” *Int. J. Emerg. Trends Technol. Comput. Sci.*, vol. 10, no. 3, pp. 67–74, May 2021.
- [15] V. Kubde, N. Chauhan, V. Jamkhandi, A. Joshi, and D. Kadam, “QuickPrint: A Web-based Printing Platform for Academic Communities,” 2025. (Own work — this project)

THANK YOU